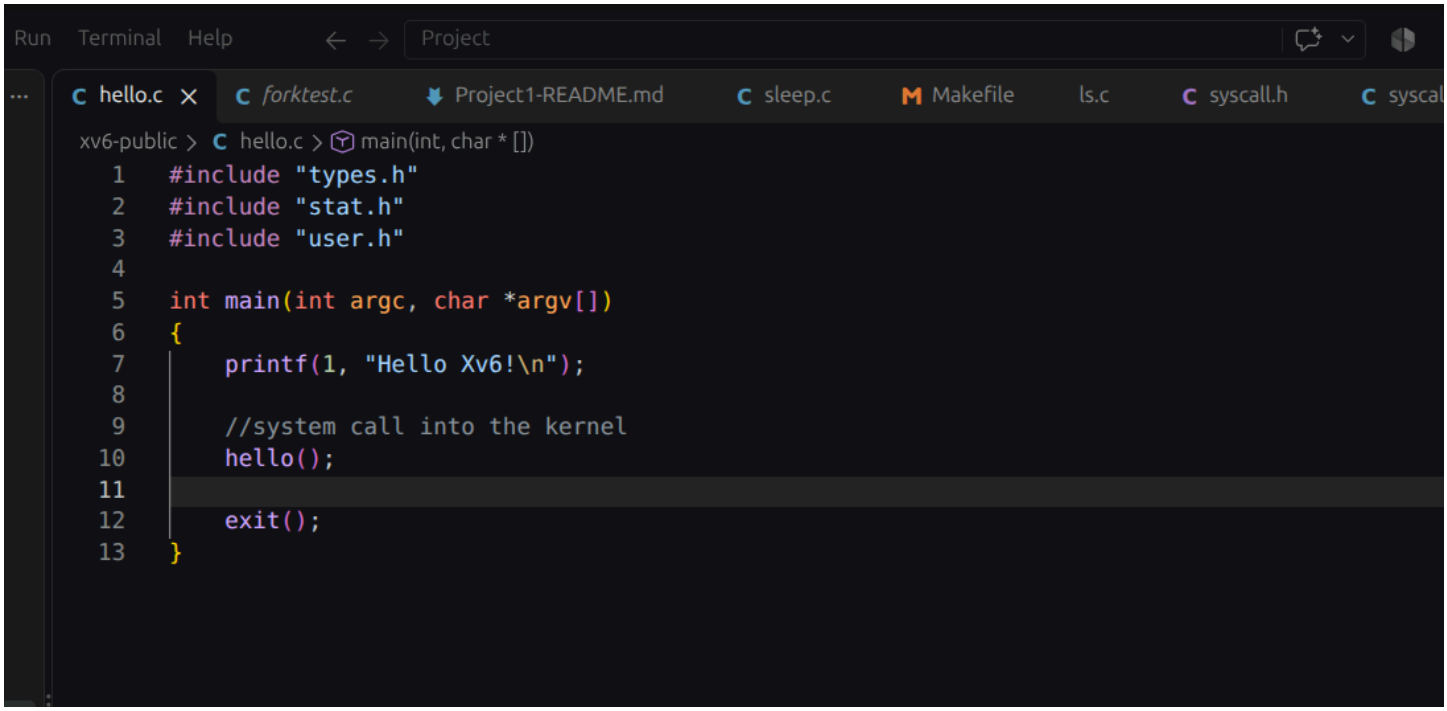
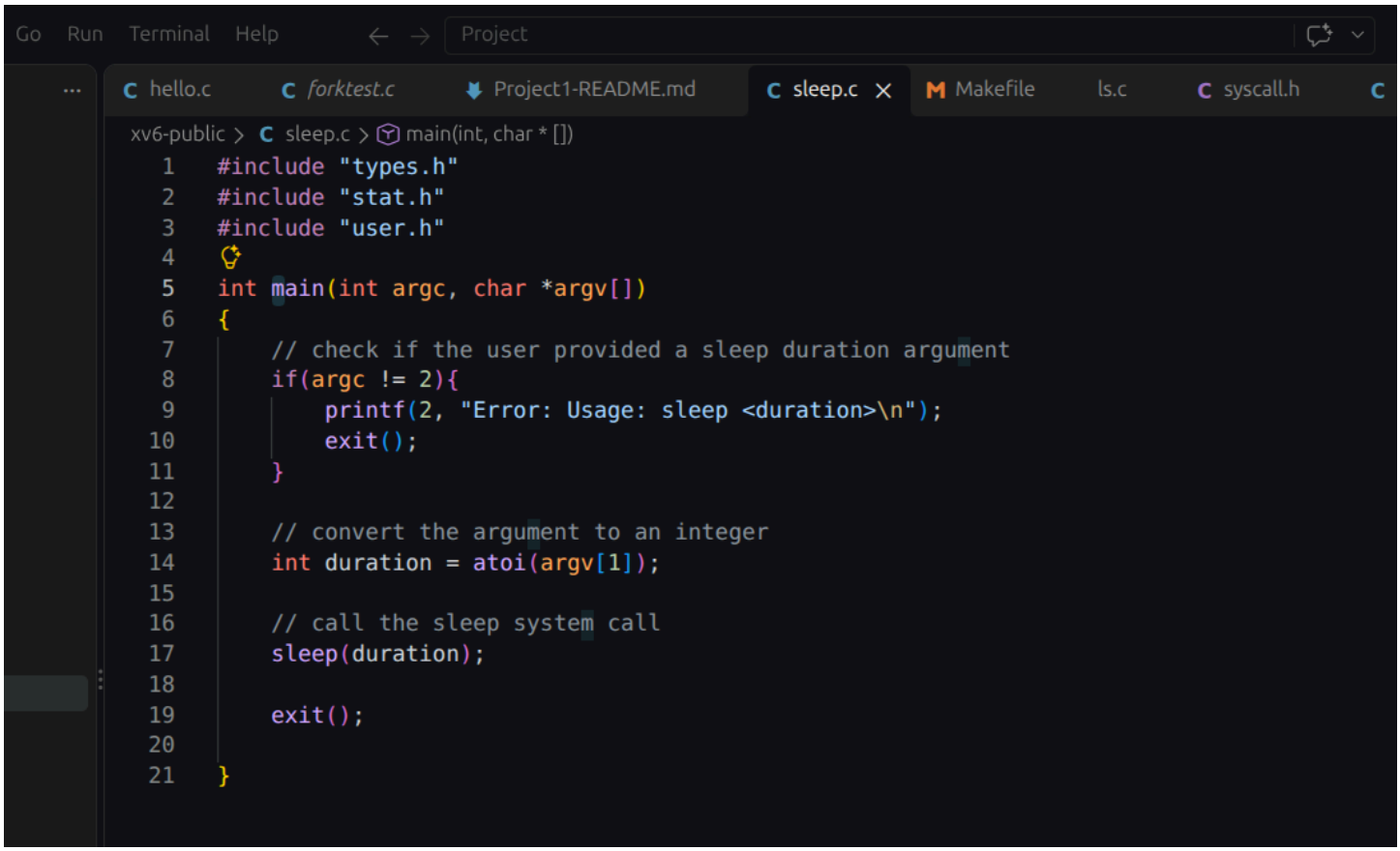
Project 1: Adding Programs and Implementing Shell Commands/System Calls in Xv6

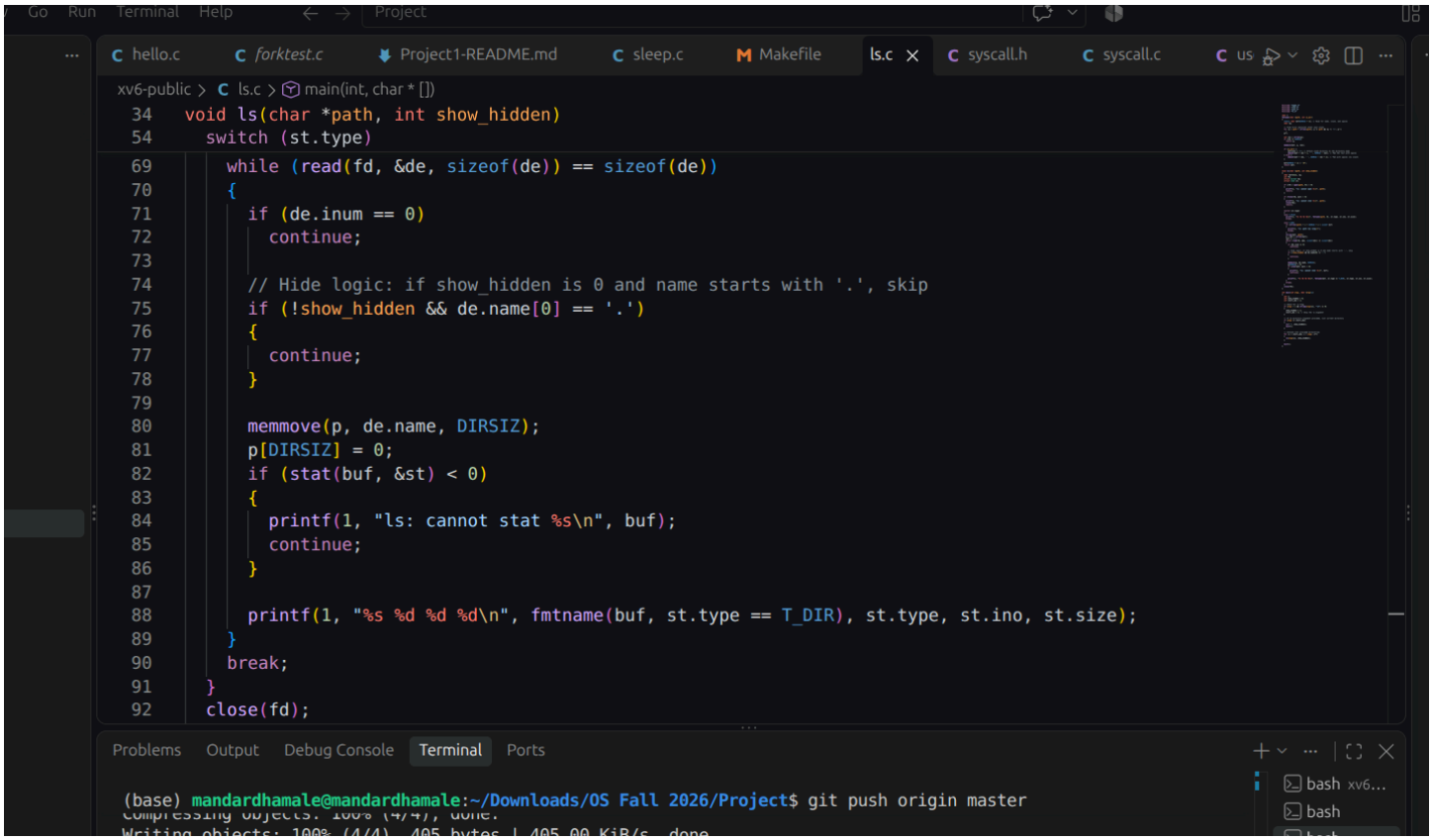
Name: Mandar Dhamale (mandardhamale@usf.edu)

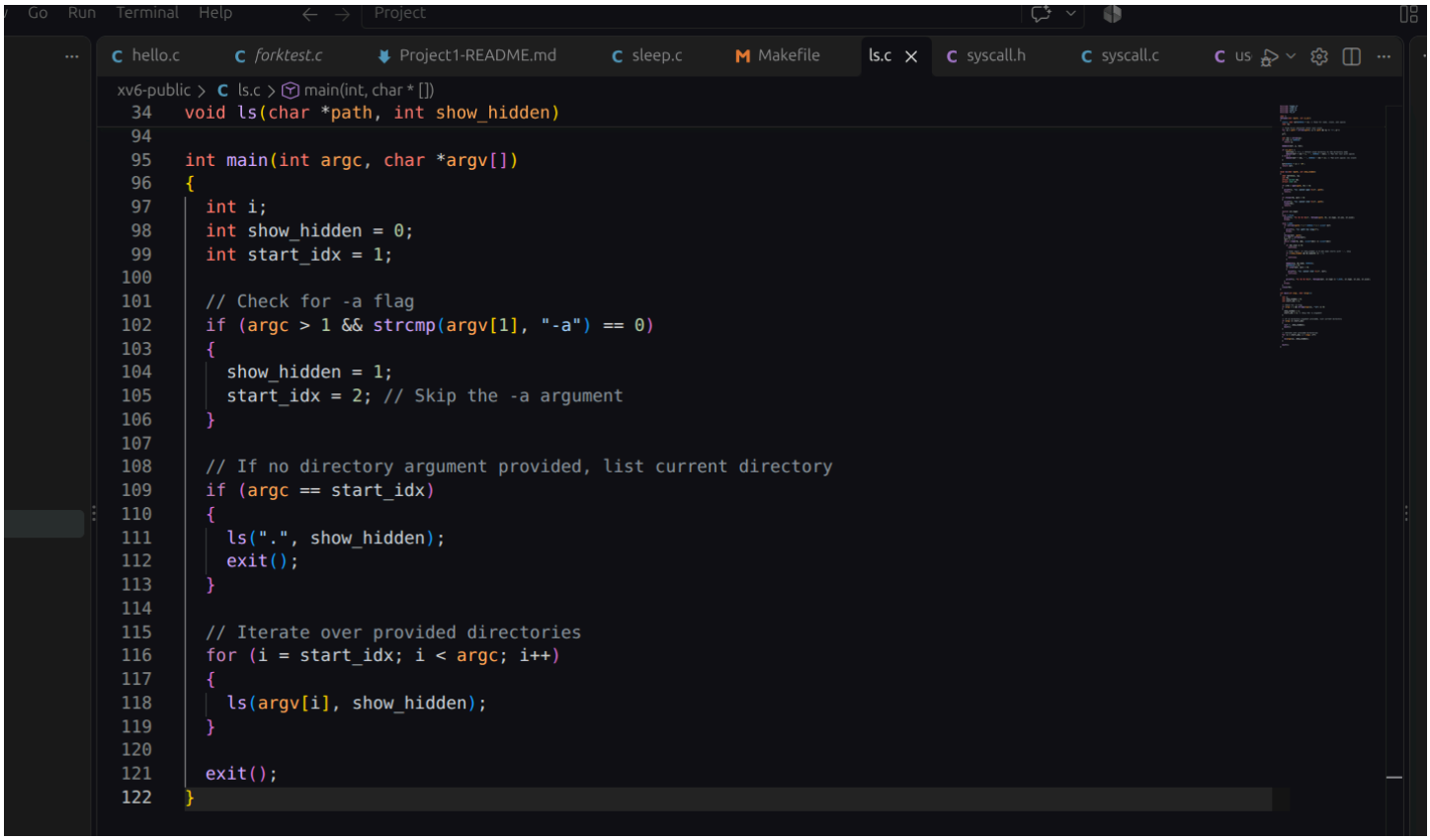
6. Verification & Screenshots

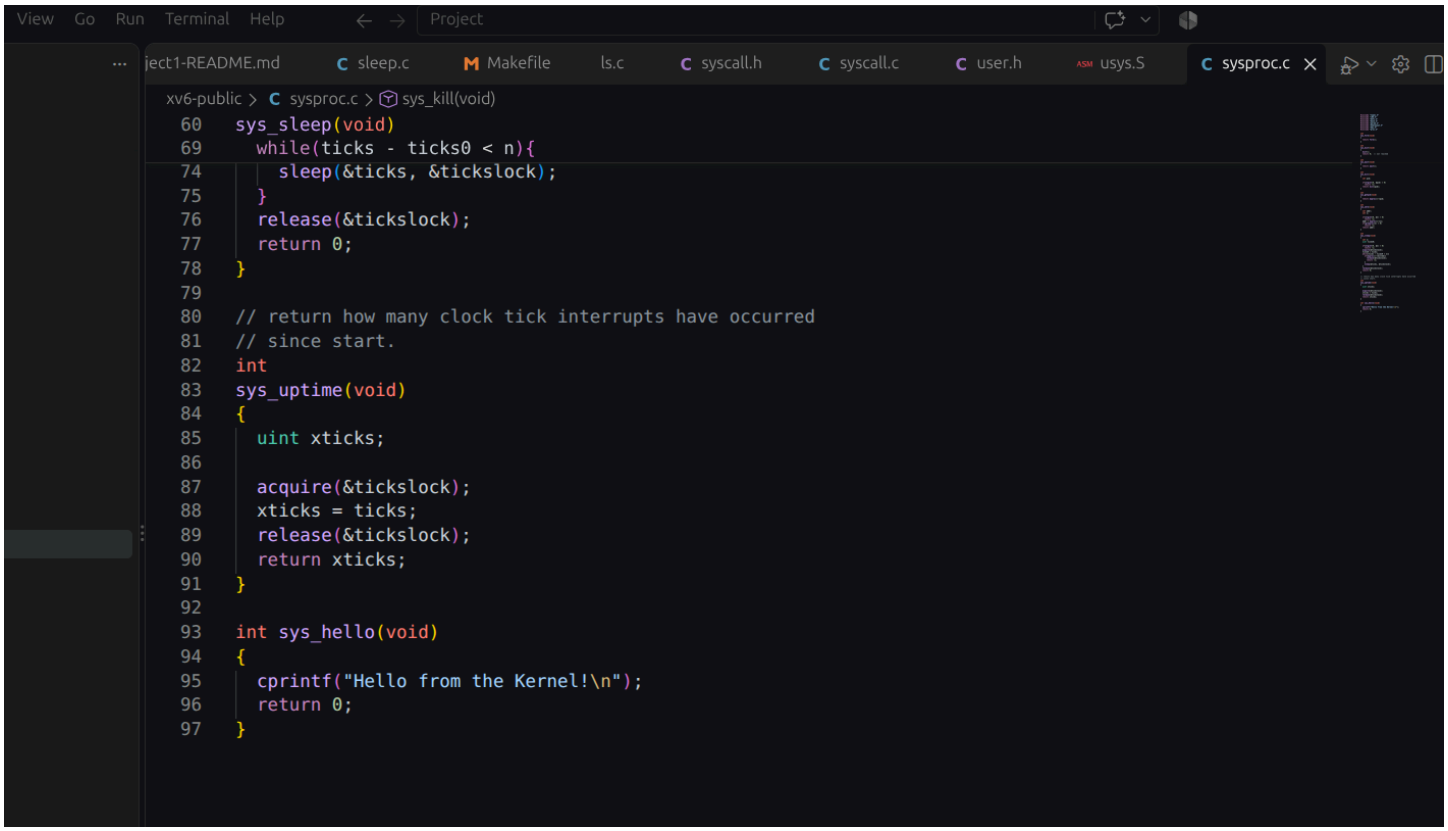
A. Source Code Implementation

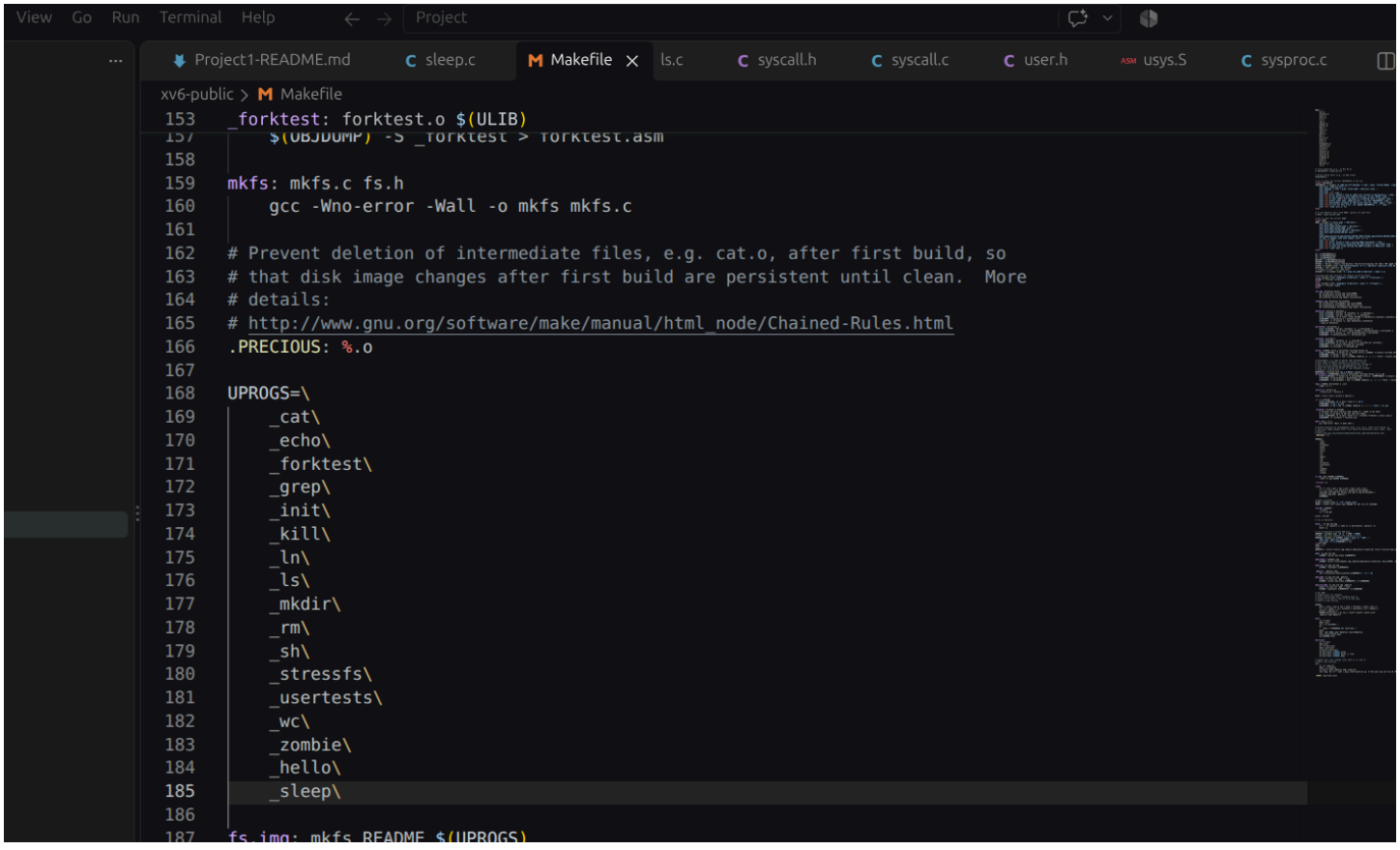




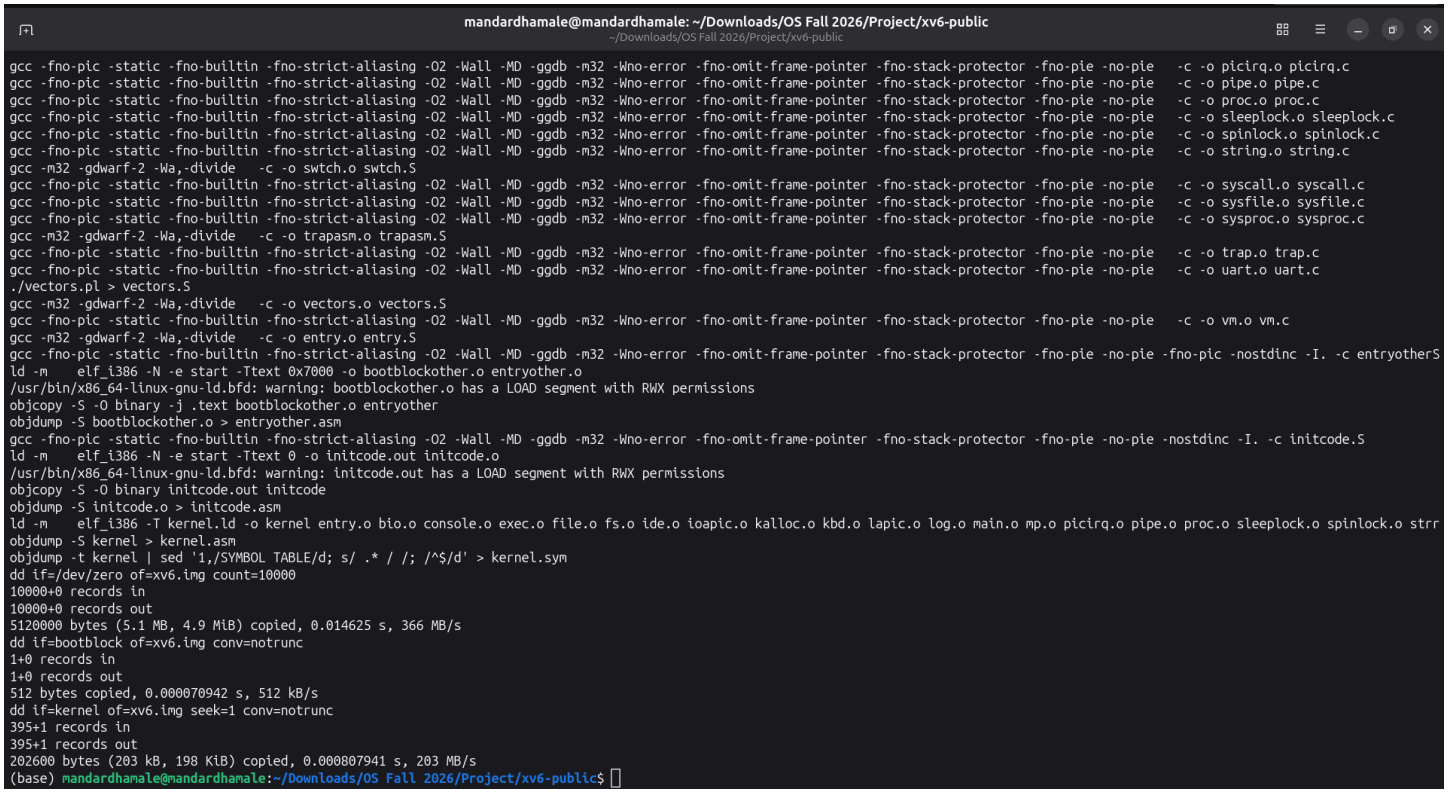

Go Run Terminal Help
Project
hello.c
forktest.c
Project1-README.md
sleep.c
Makefile
ls.c
syscall.h
syscall.c
xv6-public > ls.c > main(int, char * [])
34 void ls(char *path, int show_hidden)
54 switch (st.type)
69 while (read(fd, &de, sizeof(de)) == sizeof(de))
70 {
71 if (de.inum == 0)
72 continue;
73
74 // Hide logic: if show_hidden is 0 and name starts with '.', skip
75 if (!show_hidden && de.name[0] == '.')
76 {
77 continue;
78 }
79
80 memmove(p, de.name, DIRSIZ);
81 p[DIRSIZ] = 0;
82 if (stat(buf, &st) < 0)
83 {
84 printf(1, "ls: cannot stat %s\n", buf);
85 continue;
86 }
87
88 printf(1, "%s %d %d %d\n", fmtname(buf, st.type == T_DIR), st.type, st.ino, st.size);
89 }
90 break;
91 }
92 close(fd);
Problems
Output
Debug Console
Terminal
Ports
(base) mandardhamale@mandardhamale:~/Downloads/OS Fall 2026/Project$ git push origin master
bash xv6...
bash

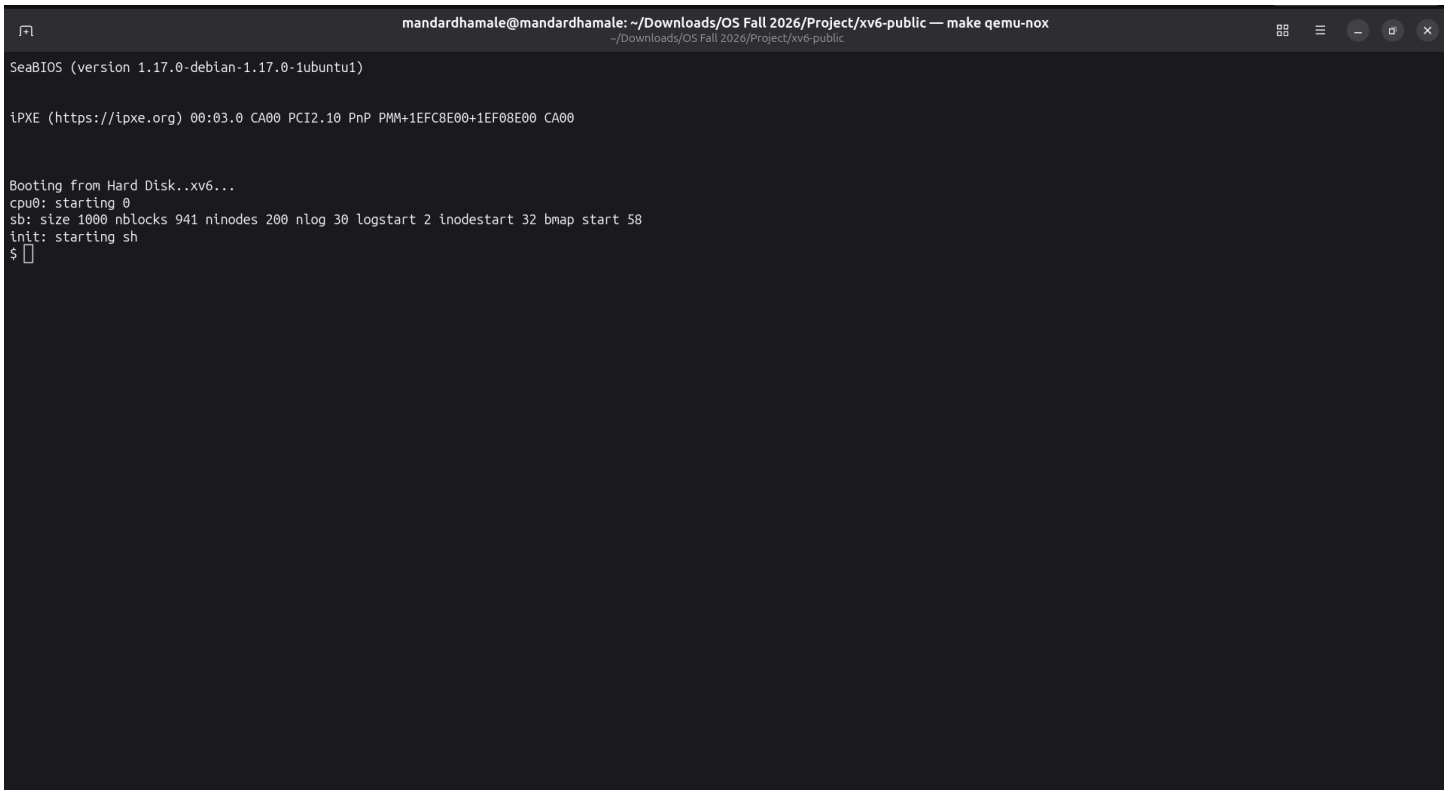

Go Run Terminal Help
Project
hello.c
forktest.c
Project1-README.md
sleep.c
Makefile
ls.c
syscall.h
syscall.c
xv6-public > ls.c > main(int, char * [])
34 void ls(char *path, int show_hidden)
94
95 int main(int argc, char *argv[])
96 {
97 int i;
98 int show_hidden = 0;
99 int start_idx = 1;
100
101 // Check for -a flag
102 if (argc > 1 && strcmp(argv[1], "-a") == 0)
103 {
104 show_hidden = 1;
105 start_idx = 2; // Skip the -a argument
106 }
107
108 // If no directory argument provided, list current directory
109 if (argc == start_idx)
110 {
111 ls(".", show_hidden);
112 exit();
113 }
114
115 // Iterate over provided directories
116 for (i = start_idx; i < argc; i++)
117 {
118 ls(argv[i], show_hidden);
119 }
120
121 exit();
122 }


View Go Run Terminal Help
Project
ject1-README.md
sleep.c
Makefile
ls.c
syscall.h
syscall.c
user.h
usys.S
sysproc.c
xv6-public > sysproc.c > sys_kill(void)
sys_sleep(void)
while(ticks - ticks0 < n){
sleep(&ticks, &tickslock);
}
release(&tickslock);
return 0;
}
// return how many clock tick interrupts have occurred
// since start.
int
sys_uptime(void)
{
uint xticks;
acquire(&tickslock);
xticks = ticks;
release(&tickslock);
return xticks;
}
int sys_hello(void)
{
cprintf("Hello from the Kernel!\n");
return 0;
}


View Go Run Terminal Help
Project
Project1-README.md
sleep.c
Makefile
ls.c
syscall.h
syscall.c
user.h
usys.S
sysproc.c
xv6-public > Makefile
_forktest: forktest.o $(ULIB)
$(OBJDUMP) -S _forktest > forktest.asm
mkfs: mkfs.c fs.h
gcc -Wno-error -Wall -o mkfs mkfs.c
Prevent deletion of intermediate files, e.g. cat.o, after first build, so
that disk image changes after first build are persistent until clean. More
details:
http://www.gnu.org/software/make/manual/html_node/Chained-Rules.html
.PRECIOUS: %.o
UPROGS=\
_cat\
_echo\
_forktest\
_grep\
_init\
_kill\
_ln\
_ls\
_mkdir\
_rm\
_sh\
_stressfs\
_usertests\
_wc\
_zombie\
_hello\
_sleep\
fs.img: mkfs README $(UPROGS)

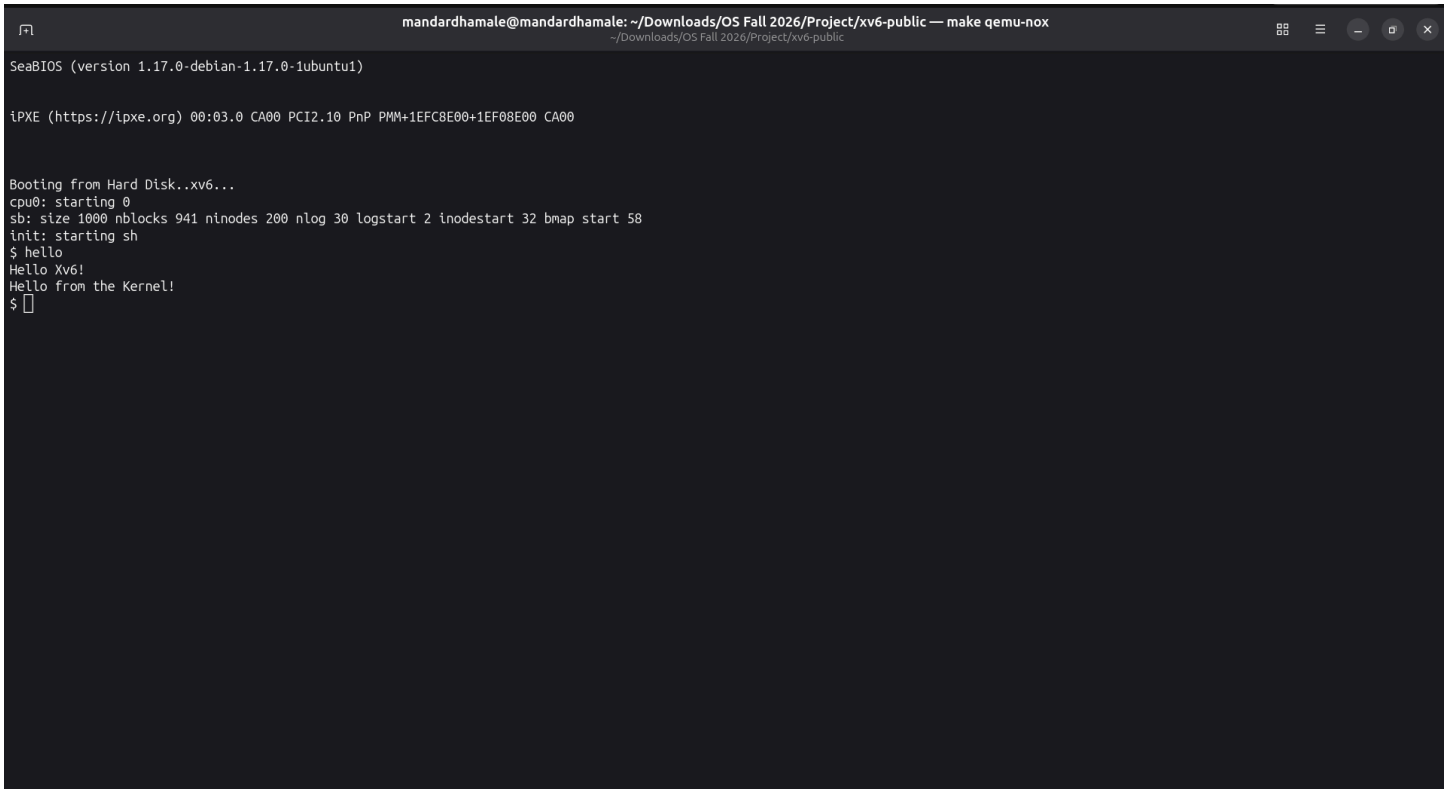
B. Build & Compilation Process



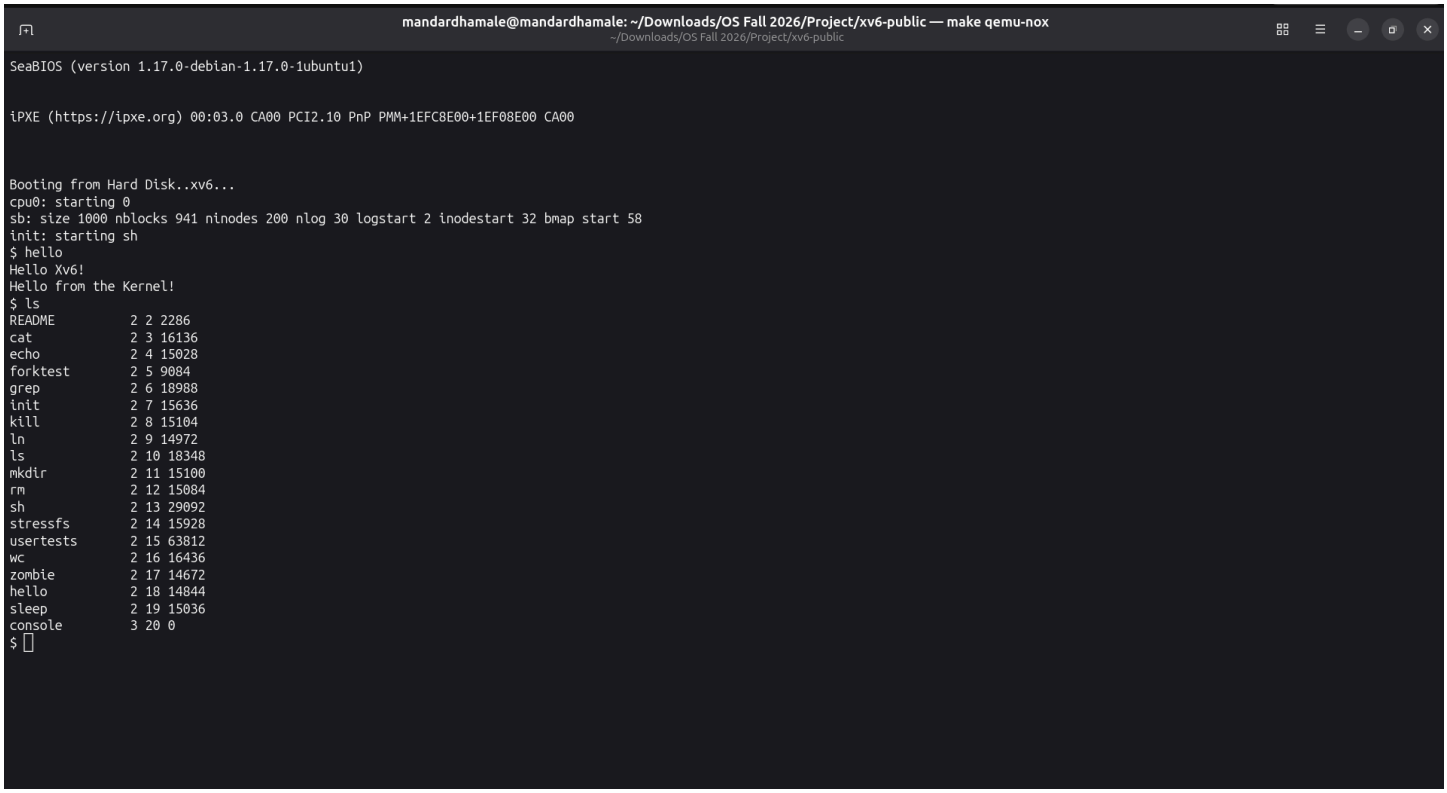


C. Execution & Functional Verification

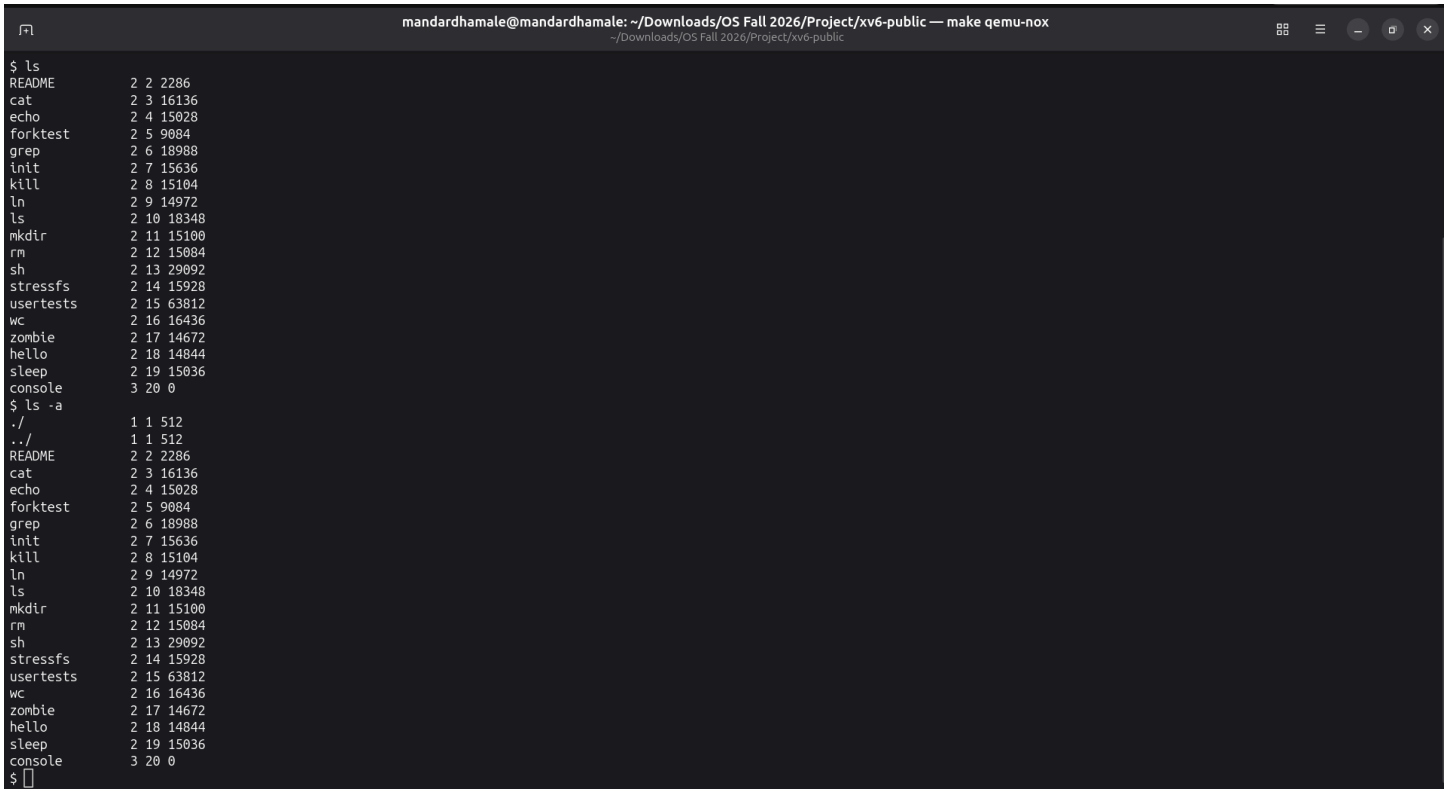
hello



ls



ls -a



sleep and sleep 100

